Low-Frequency RoPE Subspace for Near–Far Sparse Attention

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***Abstract*—Sparse attention must retain useful candidates at low selection cost. We study a near–far page-to-token cascade sharing 32 low-frequency RoPE scoring coordinates and using original-dimensional attention on selected tokens. Near–far coverage involves trade-offs, and far-page reservation need not preserve far-token capacity. Author-reported 11-task RULER 32K AVG is 58.30 and 59.29 under two Qwen3-8B page/token allocations; a separate Qwen3-32B comparison reports 50.64 for sparse attention versus 52.22 for dense attention. These descriptive comparisons neither isolate mechanisms nor establish non-inferiority. Separate historical timings show slower sparse decoding. Establishing benefit for the compact 4/128-protection design still requires matched quality and preparation-inclusive complete-request cost.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context benchmarks such as LongBench and RULER evaluate models on tasks with long input histories [9], [10]. Long-context decoding repeatedly accesses previously generated keys and values. Sparse attention limits this work by selecting a smaller token set, but selection introduces its own scoring, ranking and memory accesses. A page-to-token cascade first decides which pages may contribute tokens to refinement. The aim is to retain useful candidates within a fixed token budget at a retrieval cost below the attention work saved.

Page summaries approximate relevance with different scoring targets. Mean pooling measures an average key response, whereas sign-aware extrema bound individual unquantized token scores in the same coordinates; this bound can be loose. Neither score directly ranks pages by their summed attention weight. Following ClusterKV's token/page perspective [7], Figure 1(a) compares candidate access against a whole-page mass-rank reference on one profiled query: mean and extrema scoring miss 13 and 17 of the 32 mass-rank reference pages, respectively. These missed pages contain 59.65% and 26.46% of the reference pages' summed attention weight. Mean pooling can dilute isolated large token scores, while extrema bound each page's largest shared-space token score. Neither property determines the observed ordering of missed-page counts or full-dimensional attention mass. With enough positive-probability candidates to fill the quota, refinement cannot recover unprotected tokens from rejected pages. Summaries must therefore be assessed by candidate access as well as scoring cost.

A global TopK budget lets high-scoring candidates from one region occupy slots without guaranteeing minimum capacity for the other. In Figure 1(b), separate near and far reservations raise mean far retention from .6759 to .7207 while lowering it in 43 of 128 correlated observations. In these observations, the global arm reaches zero far retention, whereas the quota arm's minimum is .2667 (arm minima, not necessarily from the same query). This controllable coverage trade-off calls for explicit page and token capacities: reserving far pages alone need not preserve far tokens.

These observations lead to a cascade with two explicit controls. A fixed set of complete low-frequency RoPE pairs supplies the scoring coordinates for both stages; separate near–far page quotas determine regional access to token refinement. The coordinate diagnostic in Figure 1(c) motivates the compact view, although its random baseline does not preserve RoPE pairs and so cannot isolate frequency from pair structure. Original-dimensional Q/K/V are retained for final attention, so approximation enters through token selection. We also consider an optional prefill-informed far gate. The historical layer variation in panel (d) motivates this extension, but uses a different single-query statistic; the gate is disabled by default and must be assessed independently.

We formulate shared-coordinate regional selection and analyze how page and token budgets jointly constrain retrieval. The evaluation examines task quality (Q1), regional coverage (Q2) and execution cost (Q3). Qwen3 diagnostics show changes in coverage, but historical scores do not isolate either mechanism and sparse decoding is slower than dense attention. The compact 4/128-protection implementation still lacks matched quality–cost evaluation.

II. Related Work

Quest supplies sign-aware page-extremum bounds for far-page scoring, SparQ approximates attention from query-selected key coordinates, and HISA prunes the attention set hierarchically [1], [2], [3]. AsyncTLS already combines a page-to-token cascade with coordinate selection, quantization, per-query normalization followed by GQA probability aggregation, and final attention over original KV [4]. SparQ also protects a recent-token window, while HISA forces the first and last blocks into its coarse candidate set [2], [3]. We study these existing components with complete low-frequency RoPE pairs shared across both stages and separate nominal near–far L1 page quotas, distinguishing page-level eligibility from final token capacity. These page quotas do not by themselves guarantee final far-token capacity; ordinary L2 still scores the full history before masking.

Reducing attention traffic also requires considering what the selector itself reads. Token-Picker uses conservative probability estimates from partial key-bit chunks to stop further key-chunk and value transfers for pruned tokens [12]. Our selector instead ranks candidates using a fixed coordinate subset under page and token budgets; original-dimensional KV remain available for final attention. KV compression and placement methods address complementary storage and movement constraints [5], [6], [7]. A smaller scoring view in our design reduces neither original-KV capacity nor, by itself, complete-request latency.

III. Near–Far Selection in a Shared Subspace

To control both selection cost and candidate access, the cascade uses two explicit mechanisms: a shared scoring view and regional page quotas. One 32-coordinate map supports both L1 page screening and L2 token refinement, separating low-dimensional selection from original-dimensional attention (Fig. 2). The cascade operates only during decoding; prefill remains dense. L1 assigns separate page quotas to the near and far regions, so pages in one region do not compete for the other region's allocation. In the ordinary L2 path, full-history scoring precedes the retained-page mask and regional token selection. The selected token indices are merged and deduplicated with protected positions for attention using the original Q, K and V; recomputing retained logits does not recover omitted tokens.

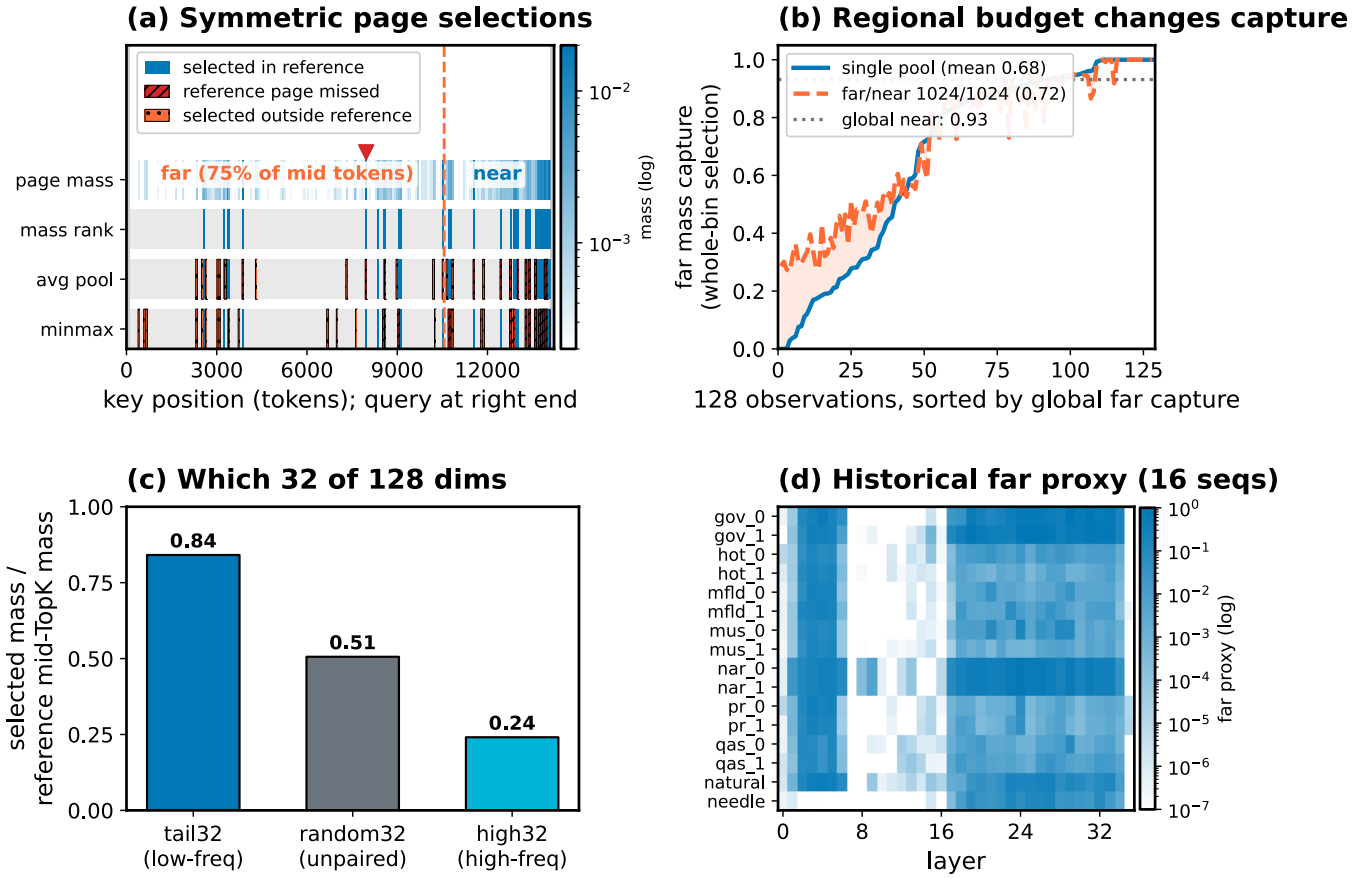


Fig. 1. Four motivating diagnostics; Qwen3-8B, 128/128 prefix/recent protection in (a)–(c). (a) MuSiQue sample 1, layer 24, query 14212. The top strip shows summed attention per page, not a selection. Below are the stored mass-rank, avg and minmax selections: 32 pages each, using 1989/1989/2048 tokens under a 2048-token cap. Blue denotes selected reference pages; red hatching denotes missed reference pages; orange dots denote selected pages outside the reference. Mass-rank is a whole-page comparison reference, not an optimal oracle, exact token TopK or DSA. Avg and minmax share 32 coordinates and head aggregation; the reference uses full-dimensional attention weights. Different regional allocations and realized token counts preclude inferring near/far specialization or task-quality ranking from these misses. (b) Four sequences contribute 128 correlated observations, ordered by global far retention. A global whole-bin cap of 2048 is compared with separate far/near caps of 1024/1024, without refill; red shading marks lower capture. Retention divides selected regional mass by full regional mass; the dotted line is global near retention. (c) Selected mass relative to full-dimensional reference mid-TopK mass; random32 does not preserve RoPE pairs. (d) One 16-by-36 historical single-query proxy matrix gives 24/36 layer medians below .01 and 12–32 low-proxy layers per sequence. It uses the last sampled query, group-logit summation before softmax and far interval $[64, t-2048)$ (S6), unlike the current gate. These are diagnostic selections and profiles, not task-quality or decode-safety results.

A. *Shared low-frequency coordinates*

Both regions and both selection stages use the same coordinate map within each layer and KV head, while each query head is scored separately. For Qwen3-8B, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. The indices retain 16 complete low-frequency split-half RoPE pairs (Supplementary Fig. S1). With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. Their slower positional rotation motivates this selection, although it does not guarantee preservation of important tokens. The compact implementation uses width-32 scoring tensors, whereas the historical quality runs used 32 active coordinates in width-128 L2 tensors (S7).

L1 aggregates selected key coordinates into page summaries for coarse ranking; L2 refines candidates using quantize–dequantize token keys. L2 uses 16 floating-point quantize–dequantize levels, not packed four-bit storage or INT4 arithmetic (S7). The two scoring views occupy separate buffers. Original KV remain full width. Compact scoring narrows dot products, but view preparation and maintenance add work without compressing the original KV cache (S7).

B. *Regional page and token budgets*

In decoding, zero-based position t includes prefill, and $S = t + 1$ counts valid cached positions through the current token, excluding padding. Let s and w denote the protected prefix and recent-window lengths. Their union and the unprotected middle are

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t - w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of $\mathcal{M}_t$; far is its complement there. Near excludes the protected recent window. Near uses fraction α of unprotected history, a one-page minimum and a downward page-aligned start; S7 specifies integer rules.

Regional page caps and token quotas constrain the cascade (Supplementary Fig. S2). Near pages use query–mean-key dot products to summarize page-level relevance; far pages use sign-aware extrema to bound unquantized shared-space token scores. This regional assignment is a heuristic. Summaries cover full physical pages, including protected boundary positions; L2 separately limits token

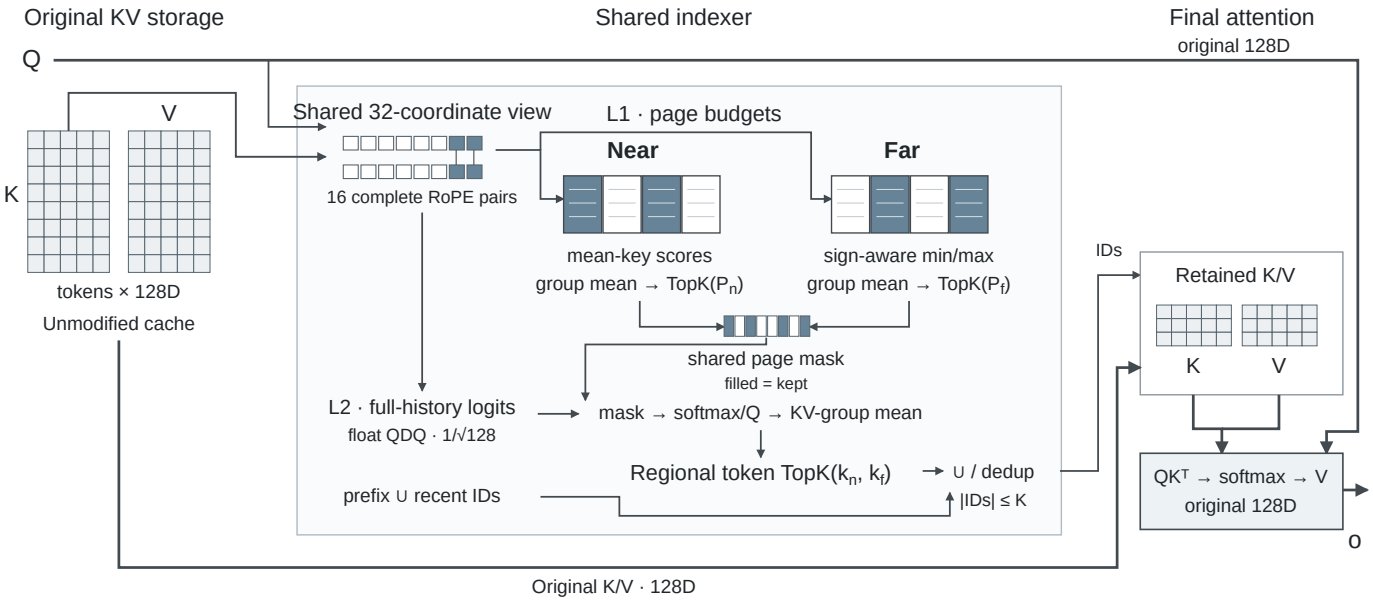


Fig. 2. Architecture of shared-subspace selection, with the optional gate inactive. A shared coordinate view feeds regional L1 page scoring and ordinary full-history L2 scoring. Page masks limit eligibility; per-query softmax and KV-group probability averaging precede regional token selection. Protected positions join deduplicated IDs within K. Near excludes the protected recent window. IDs define support over original-width K/V; physical gathering or dense masked computation depends on the backend. Original Q follows a separate path. Arrays are schematic.

regions (S7). For a nonempty page b, let $m_{b,j} = \min_{i\in b} z_{i,j}$ and $M_{b,j} = \max_{i\in b} z_{i,j}$. Quest's construction [1] gives

$$U_b(z_q) = \sum_j (z_{q,j}^+ M_{b,j} + z_{q,j}^- m_{b,j}), \tag{3}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. Each signed coordinate contribution is bounded by its chosen extremum, so $z_q^\top z_i \le U_b(z_q)$ for every unquantized shared-space key in b. This does not bound original-dimensional logits or independently quantized scores. Extrema may come from different tokens, making the bound loose enough to misrank pages and miss important tokens. Average-key scores lack this per-token bound.

For total cap $K \ge |\mathcal{P}_t|$, define the middle-token budget $K_{\text{mid}} = K - |\mathcal{P}_t|$. α controls near extent, β its share of L1 page cap K_1, and γ its nominal L2 quota from retained-page capacity. For the partitioned fixed-quota path (far active; no overrides), assume integer $K_1 \ge 1$, $\alpha > 0$, $0 < \beta \le 1$ and $\gamma \ge 0$. With page size B and sufficient regional pages/tokens, $P_n = \max(1, \text{round}(\beta K_1))$, $P_f = \max(0, K_1 - P_n)$, $k_n = \min(\lfloor \gamma P_n B \rfloor, K_{\text{mid}})$ and $k_f = \max(0, K_{\text{mid}} - k_n)$; rounding uses ties-to-even. Raising β cannot reduce the nominal near-page quota or increase the far-page quota; rounding can leave both unchanged. Thus $P_f > 0$ does not imply $k_f > 0$: page and token access are separate constraints. For $K_1 = 128$ and $\beta = .125$, L1 reserves 16 near and 112 far pages. With enough positive-probability candidates, L2 cannot recover unprotected tokens from L1-rejected pages. S7 details scarce-support and masked zero-tie exceptions.

With disjoint protection and sufficient candidates, the compact configuration ($K = 1024$, $s = 4$, $w = 128$) leaves 892 middle slots per KV head. Here $\gamma = .625$ yields $16 \cdot 64 \cdot .625 = 640$ near and 252 far slots (Supplementary Fig. S2). The partial prefix page can reselect up to four protected tokens, so 252 is nominal rather than guaranteed unprotected far coverage.

C. GQA selection and original-dimensional attention

Grouped-query attention (GQA) assigns four of Qwen3's 32 query heads to each of its eight KV heads. L1 averages their page scores before regional TopK. Ordinary L2 computes each query's full-history logits, applies the shared page mask, then softmax. Group-averaged probabilities determine regional token TopK. L2 uses scale $1/\sqrt{128}$ before softmax even with 32 scoring coordinates. Because softmax precedes group averaging, this rule generally differs from applying softmax to group-averaged logits. Width-32 scoring narrows each dot product; a smaller L1 shortlist restricts eligibility but does not reduce ordinary L2's full-history dot-product count.

For KV group h, let $\mathcal{S}_{t,h}$ be the deduplicated union of protected positions and regional winners, with $|\mathcal{S}_{t,h}| \le K$. Each query g uses this support for its own full-dimensional attention ($d = 128$):

$$o_{h,g} = \sum_{i\in\mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j\in\mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

Each query applies one softmax over the merged indices, avoiding the addition of separately normalized regional outputs [8]. Full-dimensional retained logits still use a restricted denominator, which can change the output. These indices define a shared logical support. Physical KV gathering and reuse depend on backend and length; short sequences can use a dense masked path.

D. Prefill-informed far gating

Beyond scoring and budgeting, layer variation offers a second savings opportunity. The optional gate tests whether low far-region attention during prefill can identify layers where far selection may be omitted during decoding; it is disabled by default. For each input and layer, a mean far-region attention proxy below ϵ sets the gate OFF; otherwise far selection stays active. Calibration uses a token-defined region that can differ from the page-aligned skipped region (S5). The first decode-path call freezes the decision until input reset (S5).

TABLE I
Compact design parameters and execution policy. These settings define the proposed implementation, not the historical runs. K includes unique protected tokens.

Parameter	Setting
Final token cap K	1024
Page size; L1 page cap K_1	64; 128
α, β, γ	.25, .125, .625
L1/L2 active coordinates	32/32
Prefix / recent protection	4/128
Prefill; decode layers	dense; all sparse
L2 scoring support	full history before mask

OFF skips far L1/L2 scoring and selection in the supported fixed-quota path, refilling near L2 within unchanged near pages. Removing far support can change softmax denominators and group-averaged near rankings (S5). Representation preparation, sequence-wide score buffers and normalization remain.

The gate supports one sequence; the skip path requires valid length and nonempty far history (Supplementary Fig. S3; S5). The prefill threshold does not bound decoding error; gate quality and total cost require separate evaluation.

IV. Experimental Setting

We evaluate three aspects separately: task quality (Q1), regional candidate coverage (Q2) and execution cost (Q3). Table I defines the compact Qwen3-8B design with dense prefill, sparse decoding and a 1024-token cap including four prefix and 128 recent tokens, referred to throughout as the 4/128-protection design. The historical five-task LongBench shared-subspace arm is linked to an earlier implementation: 128/128 protection and 32 active coordinates stored in zero-filled width-128 L2 keys. The comparison labels "full-dimensional page scoring" and "full-dimensional token scoring" denote reported variants whose complete execution configurations are unavailable. We retain these nominal labels without treating the runs as controlled single-stage ablations. Synthetic selector timing and model-level decode timing come from separate studies; none supplies matched quality and cost for the compact design. S6/S7 give configuration and measurement details.

Workloads and comparisons. The quality evidence comprises five LongBench tasks (question answering, summarization and code completion) [9], a historical 11-task RULER [10] 128K comparison, and author-reported 11-task RULER 32K comparisons on Qwen3-8B and Qwen3-32B. The LongBench scores compare three nominal sparse scoring configurations; dense FullKV is a separate reference in the RULER and decode-timing records. Controlled sparse comparisons must align final unique-token cap K (including protection), model, inputs and phase policy. Realized token counts and head-sharing granularity also affect KV traffic; internal page budgets need not match (S7).

Metrics. Primary quality is arithmetic task-score AVG over each fixed benchmark/length task set. Per-task gains and losses explain AVG; zero loss on every task is not required. Controlled comparisons require matched models, samples, lengths, scorers and phase policies, and aligned sparse-token budgets. Incomplete sets lack full-set AVG. Regional retention in Fig. 1(b) divides selected mass by full regional mass; Fig. 1(c) instead uses full-dimensional reference mid-TopK mass. Selector time covers a synthetic single-layer operation; synchronized TPOT measures model-level intervals after the first output token. Neither measures complete-request latency, including preparation, maintenance and input/output work. Unpaired quality and timing studies do not define a quality–latency frontier. AVG differences are descriptive, not statistical non-inferiority evidence.

V. Evaluation and Discussion

A. Q1: Does reduced scoring width preserve task quality?

The five-task means sit within a descriptive spread of 0.33 points across the three nominal scoring configurations. On HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9], shared-subspace scoring scores 47.588, the nominal full-dimensional page variant 47.910 and the nominal full-dimensional token variant 47.666 (Supplementary Fig. S4). The shared-subspace score is lower by 0.322 and 0.078 points, respectively. These differences describe the recorded configurations; they do not show that reducing scoring width preserves quality. The variants are not verified single-factor interventions, and the shared-subspace run uses the earlier 128/128-protection implementation. Consequently, neither stage's representation effect nor the compact 4/128 design's quality can be inferred from this comparison. All three configurations are sparse; dense FullKV is a different reference.

B. Q2: How do regional budgets constrain candidate access?

Regional quotas lower far retention in 43/128 correlated observations despite improving its mean (Fig. 1(b)); this is not task-quality evidence. Reported 8B 32K AVG rises 58.30→59.29 when $\beta = .125 \rightarrow .0625$ at $K_1 = 128$, $K = 1024$, protection 4/128. Pages 16/112→8/120 and tokens 640/252→320/572 change jointly. At $K_1 = 256$, $\beta = .03125$, near pages/tokens stay 8/320, far pages rise 120→248 and far tokens stay 572, yet AVG falls to 58.96; CWE rises 3.9 and FWE falls 9.67. Data/scorer records match these sparse arms; predictions remain unreplayed.

Against dense FullKV, reported 32K differences are -0.09 points (8B, different scorers) and -1.58 (32B, matching manifests; Fig. 3). Cross-machine execution and unreplayed predictions limit both. Task differences and provenance in S8 establish neither allocation causality nor non-inferiority.

Historical 128K sparse scores 23.98 versus 46.43. Reported budgets imply $k_f = 0$ despite retained far pages, but this neither verifies execution nor explains the 22.45-point loss (S6). Protection-only scores 26.09 versus 47.91 while using 256 versus 1024 tokens, jointly changing retrieval and budget.

C. Q3: Does compact scoring reduce execution cost?

Model-level decoding. The reported sparse configuration is slower than dense FullKV at every measured length (Fig. 4). Sparse TPOT medians are 87.9, 97.6 and 119.9 ms at 32K, 64K and 128K, compared with dense medians of 26.0, 39.7 and 67.3 ms. The corresponding ratios are 3.38, 2.46 and 1.78. Although the relative penalty decreases with length, absolute sparse TPOT increases. Thus these measurements show no decode-time benefit from that implementation. They include device synchronization and argmax, exclude tokenization, input transfer and text output, and use a configuration different from the compact 4/128 design. Aggregate medians cannot locate the cause of the slowdown or predict the compact implementation's performance; repetition and execution-path details are in S6.

Local selector cost. The shared-subspace selector has no lower reported aggregate time than the nominal full-dimensional page-scoring variant in the synthetic study. A separate two-round experiment (single layer, batch one, one query, 64K QKV) reports steady-state summaries of 1.3502 ms for nominal full-dimensional page scoring, 1.3598 ms for shared-subspace scoring and 1.4634 ms for nominal full-dimensional token scoring. Relative to nominal page

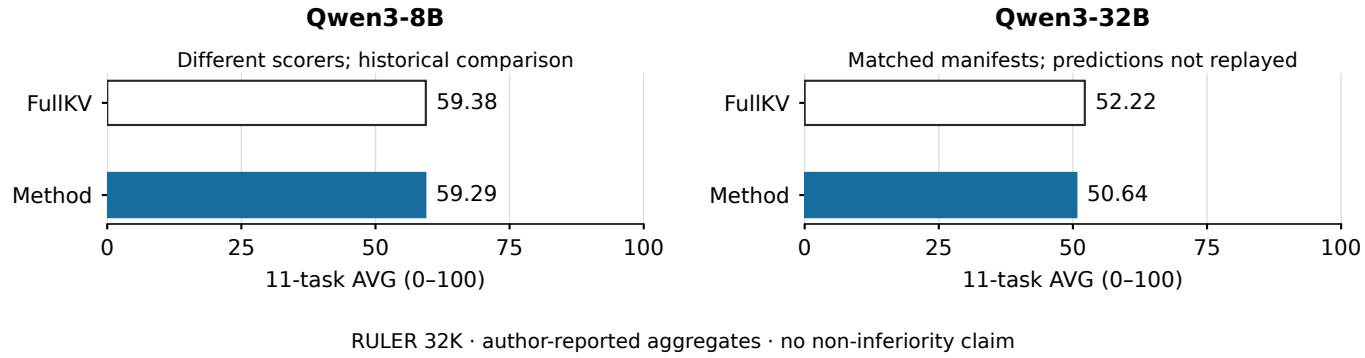


Fig. 3. Author-reported RULER 32K AVG: 11 tasks, 100 examples/task/arm. FullKV is dense; sparse arms report $K = 1024$, protection 4/128, $\beta = .0625$ (8B) or .125 (32B). Both comparisons span machines. 8B source-data hashes match but scorers differ; 32B manifests match but predictions remain unreplayed. Gate status and actual scoring dimensions are unverified. These descriptive scores establish neither non-inferiority nor matched quality–cost benefit (S8).

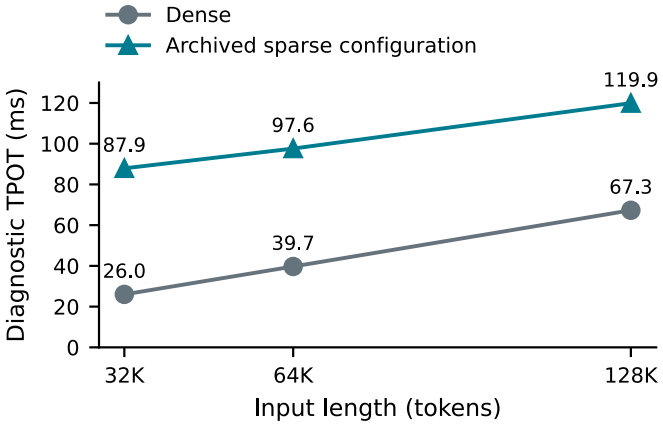


Fig. 4. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including device synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the selected 4/128 candidate. These medians do not measure complete-request E2E or establish a matched quality–cost comparison; provenance limits are in S6.

scoring, shared-subspace scoring is 0.0237 ms slower in the first round and 0.0046 ms faster in the second, showing no consistent timing advantage across rounds. Missing raw sample sequences and an unresolved sampling-protocol discrepancy prevent a reliable assessment of these small differences or a conclusion of timing equivalence (S6). These local timings do not identify a bottleneck, are not paired with LongBench quality evaluation and do not measure complete-request cost.

D. Optional gate: prefill is not a decode guarantee

Low prefill attention does not establish that far selection can be safely skipped during decoding. An archived input–layer record pairs a prefill far-attention proxy below .05 with a reported decode-tail mean of .1678 over the last five queries; this is a proxy shift, not measured task-quality loss. Separately, reported gate thresholds .05 and .10 yield arithmetic five-task AVGs of 47.876 and 47.692 against a reused 47.910 reference: decreases of .034 and .218 points. AVG is the primary quality measure. Three tasks decline in each arm, including HotpotQA by 1.05 and 1.26 points; these differences describe heterogeneity rather than impose a per-task zero-loss requirement. Neither small AVG changes nor task-bootstrap intervals containing zero establish non-inferiority. Missing complete scoring configurations, paired predictions and matched costs leave gated shared-subspace quality and net savings unverified (S6).

Design implications and limitations. Scoring and allocation rules expose two distinct constraints. Reducing scoring width narrows each dot product, but ordinary L2 still scores and normalizes the full history. Likewise, reserving far pages does not guarantee far-token capacity. These are properties of the defined rules and inspected implementation. A useful operating point additionally requires a complete-request latency reduction at a matched quality target, accounting for preparation, maintenance and selection without double-counting overlapping work. The reported historical sparse decoding is slower than dense attention, and separately measured selector times do not establish this net benefit. Matched evaluation of the compact 4/128 design remains missing; historical gate summaries also lack complete shared-subspace configuration identity. The analysis therefore identifies implementation requirements, while the combination's comparative quality–cost benefit remains unresolved.

VI. Conclusion

We studied shared low-frequency page/token scoring and regional allocation, retaining original-dimensional attention on selected tokens. The analysis separates narrower scoring from reduced execution work, and far-page reservation from far-token coverage. Regional quotas lower far retention in some Qwen3 diagnostic observations. Author-reported Qwen3-8B RULER 32K AVG (11 tasks) is 0.99 points higher at $\beta = .0625$ than at $\beta = .125$, with page and token quotas both changed. The larger nominal far-page-budget arm reports AVG 58.96, 0.33 below the $\beta = .0625$ arm's 59.29; these two arms have the same nominal token quotas. A separate Qwen3-32B sparse configuration trails dense attention by 1.58 points. These descriptive results neither isolate mechanisms nor establish non-inferiority. Historical 128K quality loss and slower sparse decoding remain limitations of their reported configurations. Establishing benefit for the compact 4/128 design still requires matched quality and complete-request cost, including preparation, maintenance and selection.

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